1

$\varepsilon = 19\%$

$\rho = 14.3\ g/cm^3$ $\sigma_f = 587 \times 10^{-24}\ cm$ nitrogen mass $= 14$

$\phi = 4 \times 10^{12}\ n/cm^2\text{-}s$ $E_f = 170\ MeV \cdot 1.604 \times 10^{-9}\ J/eV$

$Q = \underline{E_f} \times N_f \times \sigma_f \times \Phi$

molar mass

$\cancel{N_f} = (.19 \times 235 + .81 \times 238) + 14 =$

$44.65 + 192.78 + 14 =$

$$\begin{array}{r} 192.78 \\ +\ 44.65 \\ +\ 14.00 \\ \hline 251.43 \end{array} \qquad \begin{array}{r} 235 \\ \times\ .19 \\ \hline 2115 \\ 2350 \\ \hline 44.65 \end{array} \qquad \begin{array}{r} 236 \\ \times\ .81 \\ \hline 238 \\ 19040 \\ \hline 192.78 \end{array}$$

$Q = \cancel{N_f} \times 587 \times 10^{-24}\ cm \times 4 \times 10^{12} \frac{n}{cm^2 s} \times 170\ MeV \cdot 1.604 \times 10^{-9}\ J/eV$

$\cancel{N_f = \frac{1\ mol}{251.438} \cdot \frac{14.34}{cm^3} \cdot \frac{1U}{UN} \cdot .19}$

$N_f = \frac{1\ mol}{251.437} \cdot \frac{6.022 \times 10^{23}}{mol} \cdot \frac{1U}{UN} \cdot .19 \cdot \frac{14.34}{cm^3}$

Forgot calculator

(2)

$r = .1$

Cladding $K = .18$ W/cmK

Fuel $K = .04$ W/cmK

Coolant $h = 1.5$ W/cm²K

$Q = 350$ W/cm³

$R_f = .45$ cm

$t_g = 80\,\mu m$

$t_{clad} = .06$ cm

$T_{cool} = 540$ K

$$\begin{array}{r} 4 \\ \times .04 \\ \hline .16 \end{array}$$

350

$16\overline{)35000}$ 2124

$$\begin{array}{r} 124 \\ \times .45 \\ \hline 120 \\ 960 \\ \hline 10.80 \end{array}$$

$h_{clad} = \frac{K_{clad}}{t_{clad}} = \frac{.18}{.06}$

$T_0 - T_s = \frac{Q}{4K} R_f^2 \Rightarrow \left(\frac{350}{4(.04)}\right) .45^2 \Rightarrow 21.24 \cdot .45$

$.06\overline{)18} \Rightarrow .02$

$h_{cool} = 1.5$

$h_{gap} = \frac{K_{gap}}{t_{gap}} = \frac{16\times10^{-6} \times 6^{79.64}}{80\,\mu m}$

$T_s - T_{IC} = \frac{Q}{2h_{gap}} R_f$

$T_{IC} - T_{OC} = \frac{Q}{2h_{clad}} R_f \Rightarrow T_{IC} - T_{OC} = \frac{350}{(2)(.02)} = 874.10 R_f$

$4\overline{)35000}$ 8740

$$\begin{array}{r} 2 \\ \times .02 \\ \hline .04 \end{array}$$

$874.10 \cdot .45 = 393.35$

$$\begin{array}{r} 874.10 \\ \times .45 \\ \hline 437050 \\ 3496400 \\ \hline 393345.0 \end{array}$$

$T_{OC} - T_{cool} = \frac{Q}{2h_{cool}} R_f = \frac{350}{2(1.5)} \cdot .45 = 52.290$

$T_{OC} = 52.290 + 540K$

$T_{OC} = 592.29K$

$T_{IC} = 679.69K$

$T_{OC} = 592.290$

$3\overline{)350}$ 116.2

$2 \times 1.5 = 3$

$T_s - T_{IC} = \frac{Q}{2h_{gap}} R_f$

$T_{IC} = T_{OC} + 87.4$

$T_s = \frac{Q}{2h_{gap}} R_f + 679.69$

$$T_{IC} = \begin{array}{r} 592.29 \\ +\ 87.40 \\ \hline 679.69 \end{array}$$

$$\begin{array}{r} 116.2 \\ \times .45 \\ \hline 5810 \\ 46480 \\ \hline 522.90 \end{array}$$

$T_0 = \frac{Q}{4K} R_f^2 + T_s$

$K_{gap} = 16\times10^{-6} \times T^{.79}$

$T(.1cm) = \frac{Q(R^2 - r^2)}{4k} + T_s$

$K_{gap} = 16\times10^{-6} \times 679.69K^{.79}$

$T(.1cm) = \frac{350(.45^2 - .1^2)}{4 \cdot .04} + T_s$

Forgot Calculator

(3)

Rod length = 3.6m $c_p = 4200 \frac{J}{kg K}$ $T_{in} = 580K$

$LHR^0 = 400 W/cm$ $\dot{m} = \frac{.1 \ kg}{s \ rod}$

$\gamma = 1.25$ $z_0 = 3.6/2 = 1.8m$

LHR

LHR at $z = 2.1m$? T_{cool}?

$$LHR\left(\frac{z}{z_0}\right) = LHR^0 \cos\left[\frac{\pi}{2\gamma}\left(\frac{z}{z_0} - 1\right)\right]$$

$$LHR = 400 \cos\left[\frac{\pi}{2 \cdot 1.25}\left(\frac{2.1}{1.8} - 1\right)\right]$$

$$T_{cool}(z) - T_{cool}^{in} = \frac{2\gamma}{\pi} \frac{z_0 \times LHR^0}{\dot{m} c_p}\left\{\sin\frac{\pi}{2\gamma} + \sin\left[\frac{\pi}{2\gamma}\left(\frac{z}{z_0} - 1\right)\right]\right\}$$

$$T_{cool}(2.1) = \frac{2 \cdot 1.25}{\pi} \frac{1.8 \times 400}{.1 \cdot 4200}\left\{\sin\frac{\pi}{2 \cdot 1.25}\left(\frac{2.1}{1.8} - 1\right)\right\}$$

Forgot Calculator

(4)

$T_{furn} = 1200\ K$

$FIMA = 5\%$

$$\begin{array}{r} 200 \\ \times\ .05 \\ \hline 10.00 \end{array}$$

$A = 3.8 + 200 \times .05 = 13.8$

$B = .0217$

$$\begin{array}{r} 1200 \\ \times .0217 \\ \hline 8400 \\ 12000 \\ 240000 \\ 000 \\ \hline 26.0400 \end{array}$$

$5\% \Rightarrow K_{ox} = \frac{1}{A+BT} \Rightarrow \frac{1}{13.8 + .0217(1200)}$

$= 26.04 + \cancel{13.8} = \frac{1}{39.12}$

$$\begin{array}{r} 26.04 \\ +13.08 \\ \hline \end{array}$$

fresh $\Rightarrow$ $K_{ox} = \frac{1}{3.8 + 26.04} \Rightarrow \frac{1}{29.84}$

Forgot Calculator

(8) Smear density is the ratio of fuel volume to cladding volume. A reduced smear density is needed to minimize adverse mechanical effects from stress and strain of fuel swelling

(9) Fission Gas causing swelling and reducing density. Temperature increases decrease thermal conductivity due to the temperature dependence of thermal conductivity, driven from phonon interactions. Low frequency phonons dominate thermal conductivity.

(10) Cladding prevents corrosion of fuel from coolant, contains fission products, maintains fuel geometry, transfers heat to coolant

Zirconium is a suitable clad material because of

- cost
- abundance
- neutron transparent
- good thermal conductivity
- low corrosion in water
- little FCCI with UO_2
- manufacturability
- mechanical properties

(11) fuel, gap, cladding, coolant

(12) heat generation, performance during & after accident conditions, causes of down time

(13)

Finite difference	Finite volume	Finite element
- easy to solve	- can model any geometry	- can model any BC

(14)

Positive	negative
1 - little to no FCCI	1 - relatively bad thermal conductivity
2 - maintains stable phase to melt	2 - non-negligible swelling

(15) implicit time integration takes the derivative of the current time step plus the value of the current time step to predict the next time step

explicit takes the derivative of the next time step plus the value of the current time step to predict the next time step values